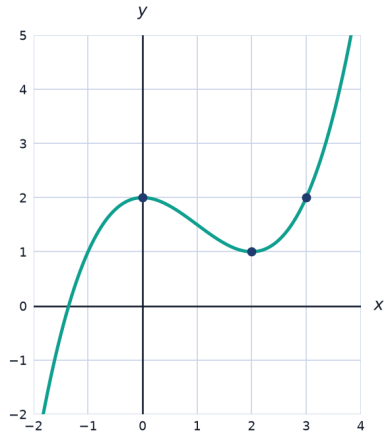
The laws of exponents and scientific notation

Core Concepts: The laws of exponents and scientific notation

1 For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(2)$.



a. 14

b. 19

c. 10

d. 18

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(3)$.

a. 19

b. 24

c. 15

d. 23

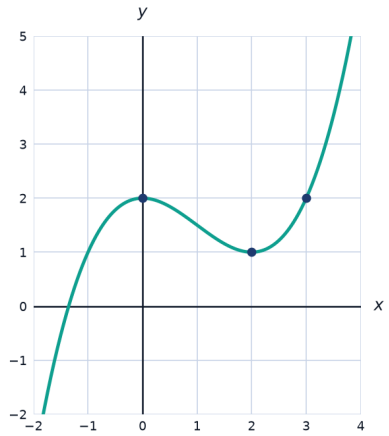
5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$



6 A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(4)$.

a. 24

b. 29

c. 20

d. 28

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

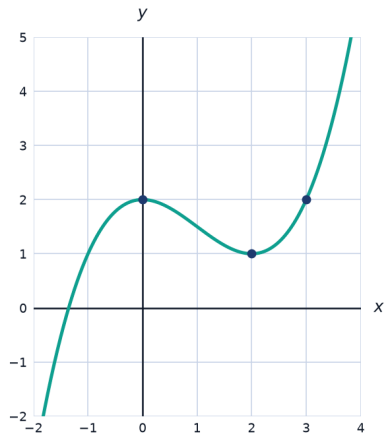
9 A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

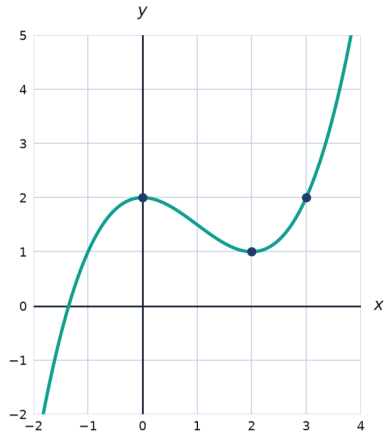
d. 25.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(5)$.



a. 29

b. 34

c. 25

d. 33

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

12 A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

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